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Sang Hyeok Im (임상혁)

Curriculum Vitae

Department of Physics and Astronomy, Seoul National University, 1, Gwanak-ro, Gwanak-gu, Seoul, Republic of Korea — sanghyeok.im97@gmail.com

EDUCATION

Ph.D student in Astronomy Program (2021.02 - present)

Department of Physics and Astronomy, Seoul National University, 1, Gwanak-ro, Gwanak-gu, Seoul, Republic of Korea

B.S. in Physics (2015.03 - 2021.02; double major)

Kyungpook National University, 80, Daehak-ro, Buk-gu, Daegu, 41566, Republic of Korea

B.S. in Astronomy and Atmospheric Sciences (2015.03 - 2021.02; double major)

Kyungpook National University, 80, Daehak-ro, Buk-gu, Daegu, 41566, Republic of Korea

PUBLICATIONS

Refereed Journal Article

First Author

1. Testing Ly α Emitters and Lyman-Break Galaxies as Tracers of Large-Scale Structures at High Redshifts
Sang Hyeok Im, Ho Seong Hwang, et al.,
2024, ApJ, 972, 196 (NASA ADS)
Highlighted as AAS Nova¹ (see here)

Co-Author

1. ODIN: Searching for LyC emission from Lyman- α emitters at $z = 4.5$ in the E-COSMOS and XMM-LSS fields
Eunsuk Seo et al., (**Sang Hyeok Im**)
2025, accepted for publication in ApJ (arXiv:2603.15825)
2. ODIN: Clustering Properties of Ly α Blobs at $z \sim 2.4$ and 3.1
Byeongha Moon et al., (**Sang Hyeok Im**)
2025, Submitted to ApJ (arXiv:2601.01991)
3. ODIN: A New Lyman Alpha Blob Selection Method, Sample, and Statistical Analysis at $z \sim 3.1$
Byeongha Moon et al., (**Sang Hyeok Im**)
2025, ApJ, 998, 345 (NASA ADS)
4. ODIN: Characterizing the Three-dimensional Structure of Two Protocluster Complexes at $z=3.1$
Vandana Ramakrishnan et al., (**Sang Hyeok Im**)
2025, submitted to ApJ (arXiv:2511.11826)
5. ODIN: Star Formation Histories Reveal Formative Starbursts Experienced by Ly α -emitting Galaxies at Cosmic Noon
Nicole M. Firestone et al., (**Sang Hyeok Im**)
2025, ApJL, 986, 8 (NASA ADS)
6. ODIN: High Clustering Strength of Protoclusters at Cosmic Noon
Vandana Ramakrishnan et al., (**Sang Hyeok Im**)
2025, ApJ, 982, 74 (NASA ADS)
7. ODIN: Identifying Protoclusters and Cosmic Filaments Traced by Ly α -emitting Galaxies
Vandana Ramakrishnan et al., (**Sang Hyeok Im**)

¹AAS Nova is the research highlights from the journals of the American Astronomical Society.

2024, ApJ, 977 119 (NASA ADS)

8. ODIN: Improved Narrowband Ly α Emitter Selection Techniques for $z = 2.4, 3.1$, and 4.5
Nicole M. Firestone et al., (**Sang Hyeok Im**)
2024, ApJ, 974, 217 (NASA ADS)
9. The One-hundred-deg² DECam Imaging in Narrowbands (ODIN): Survey Design and Science Goals
Kyoung-Soo Lee et al., (**Sang Hyeok Im**)
2024, ApJ, 962, 36 (NASA ADS)
10. ODIN: Where Do Ly α Blobs Live? Contextualizing Blob Environments within the Large-Scale Structure
Vandana Ramakrishnan, Byeongha Moon, **Sang Hyeok Im** et al.,
2023, ApJ, 951, 119 (NASA ADS)

GRANTS AND FELLOWSHIPS

CIAS Student Fellow (2021.09 - 2026.02)

- o Institutional Location: Korea Institute for Advanced Study, 85 Hoegi-ro, Dongdaemun-gu, Seoul 02455, Republic of Korea
- o Project Title: A 100 Deg² DECam Narrow-band Survey for the LSST Era: Tracing the Largest Cosmic Structures in the Distant Universe

PRESENTATIONS

Oral Presentations

1. **110th KAS Fall Meeting**
Date: 2024 Oct. 16 - 18
Location: Gangneung-si, Korea
Title: Testing LAEs and LBGs as Tracers of Large-Scale Structures at High Redshifts
2. **K-GMT Science Program Users Meeting 2024**
Date: 2024 Aug. 21 - 23
Location: KASI, Daejeon, Korea
Title: Ongoing Morphological Transformation from Elliptical to Spiral Galaxies Induced by Gas Accretion from Neighboring Galaxies
3. **9th Galaxy Evolution Workshop**
Date: 2023 Feb. 20 - 23
Location: Kyoto University, Japan
Title: Testing Lyman alpha emitters and Lyman-break galaxies as tracers of large-scale structures in the high-redshift universe

Poster Presentations

1. **ELT Science in Light of JWST**
Date: 2024 June 3 - 7
Location: Tohoku University, Sendai, Japan
Title: Large-scale structures in the universe traced by Lyman alpha emitters and Lyman-break galaxies
2. **108th KAS Fall Meeting**
Date: 2023 Oct. 18 - 20
Location: Ramada Plaza Jeju, Jeju, Republic of Korea
Title: Testing Lyman alpha emitters and Lyman-break galaxies as tracers of the large-scale structures in the high-z universe

3. **XXXIth General Assembly International Astronomical Union**

Date: 2022 Aug. 2 - 11

Location: BEXCO, Busan, Republic of Korea

Title: Lyman alpha emitters and Lyman-break galaxies as tracers of large-scale structures in a simulated universe

RESEARCH INTERESTS

1. Large-scale structures of the universe
2. Observational cosmology
3. Environmental effects on galaxy properties
4. Formation and evolution of proto-clusters